1. Clinical Question & Core Hypothesis

Clinical Question

The Ontologist highlighted that some MIBC tumors with high systemic neoantigen load (bTMB) and inflammatory signatures still respond variably to immunotherapy; why does high mutational load fail to translate into effective anti-tumor immunity in a subset of patients?

Working Hypothesis

In muscle-invasive bladder cancer (MIBC), S100A9-driven myeloid inflammation establishes an immunosuppressive tumor microenvironment that functionally suppresses tumor-infiltrating lymphocytes, resulting in reduced TIL activity and low TIL density despite high bTMB. Consequently, MIBC tumors with high bTMB and high S100A9-driven inflammation accompanied by low TIL density will show poorer immunotherapy responses than tumors where high bTMB coexists with preserved TIL density and activity.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	bTMB, S100A9
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

bTMB: bTMB) and inflammatory signatures still respond variably to immunotherapy; why does high mutational load fail to translate into effective anti-tumor immunity in a subset of patients? (2) Tier 1 paths used: I built primarily on the bTMB ←→ S100A9 → immune-suppression axis because it provides a clear mechanistic route by which myeloid-driven inflammation can blunt T-cell function and thus modulate the impact of neoantigen load.

S100A9: S100A9 → immune-suppression axis because it provides a clear mechanistic route by which myeloid-driven inflammation can blunt T-cell function and thus modulate the impact of neoantigen load.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	PD-L1	Supporting modifier from knowledge graph

Secondary	SDC2	Supporting modifier from knowledge graph
Secondary	TFPI2	Supporting modifier from knowledge graph
Secondary	TP53	Supporting modifier from knowledge graph
Secondary	MSI	Supporting modifier from knowledge graph
Secondary	ER status	Supporting modifier from knowledge graph
Secondary	HPV status	Supporting modifier from knowledge graph
Secondary	AI-based biomarkers	Supporting modifier from knowledge graph
Secondary	PSA	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: with a single imaging feature (TIL density) directly tests whether a high neoantigen load can be negated by myeloid-driven immunosuppression, thereby explaining variability in response to immunotherapy within MIBC.

Prediction 2: as modifiers to refine risk strata, while keeping them outside the primary hypothesis. Endpoints: therapy_response (RECIST/iRECIST or immune-related response criteria), progression-free survival, and overall survival; TIL density/function as mechanistic readouts; replication across independent cohorts.

Prediction 3: In retrospective cohort data, the prognostic benefit of the imaging feature will be modified by the anchor gene expression level (significant interaction term).

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- PD-L1: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- SDC2: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.